

Efficient Harmonic Balance Analysis of Waveguide Devices with Nonlinear Dielectrics

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Abstract—The computation of high order harmonic components of the time-harmonic field originated in waveguide structures containing nonlinear dielectrics is addressed exploiting harmonic balance and finite element methods. The solution of the nonlinear system is handled either via Picard or relaxed iterations. Being the nonlinear loop particularly time demanding, a domain decomposition method is proposed to speed-up the system solution by restricting the iteration to a small subdomain.

Index Terms—Harmonic analysis, finite element methods, domain decomposition method, nonlinear media, waveguide filters.

I. INTRODUCTION

TODAY's microwave and millimeter wave passive devices design requires, when high power densities are involved, the modeling of material nonlinear properties to accurately predict the steady state electromagnetic field.

Among the available techniques, the most immediate would be an explicit time-domain approach, which indeed often results to be computationally prohibitive [1]. An alternative is the time-harmonic approach, which can be based on the finite element (FE) solution of a frequency domain partial differential equation (PDE), obtaining the field quantities at a given frequency. Unfortunately, a single frequency is not sufficient to include all relevant phenomena, high order harmonics being generated from the interaction of the electromagnetic field with the nonlinear materials. Incorporating a larger set of harmonics in the FE formulation by using the harmonic balance finite element (HBFE) method [2], [3], also known as multiharmonic finite element method [4]–[6], provides a fast and accurate solution to the problem as long as the harmonics required to effectively represent the field are not excessive.

The nonlinearity of the PDE is tackled by an iterative solution of a spatially and spectrally coupled large-scale linear system in which the nonlinear material properties are computed at each iteration on the basis of the previous iterations field values. Iteration stops when the computed solution converge, within a prescribed tolerance, to the sought solution. An algebraic domain decomposition (DD) technique as proposed in [7] improves computational timings, especially when the region occupied by nonlinear materials is rather small. But in [7] no harmonic balance was present and hence only the fundamental frequency was evaluated.

In this letter, we expand [7] by implementing a full harmonic balance domain decomposition finite element (HBDDFE)

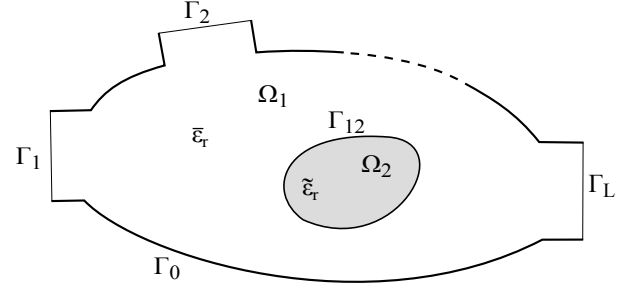


Fig. 1. Generic multiport device where the total domain Ω is subdivided into a large, linear subdomain Ω_1 and a smaller subdomain Ω_2 .

method. To prove the capability and efficiency of the method, the analysis of a waveguide filter containing dielectric materials which present a nonlinear isotropic permittivity is shown.

In following section II, the HBDDFE formulation is presented. In section III, simulation results for a broadband E-plane bandpass filter with a dielectric slab exhibiting a Kerr-type nonlinearity are shown, and the performances of the HBDDFE are discussed.

II. FORMULATION

The formulation given in this section holds for a generic multiport device as shown in Fig. 1. The domain Ω , with boundary Γ , can be either defined on $\mathbb{R}^2$ or $\mathbb{R}^3$. For the sake of simplicity, the examples presented in the next section are analyzed, without loss of generality, in $\mathbb{R}^2$. Ω is divided into a large part, Ω_1 , comprising only linear media, and a small part, Ω_2 , comprising all nonlinear media, with $\Omega = \Omega_1 \cup \Omega_2$ and $\Omega_1 \cap \Omega_2 = \emptyset$.

The permittivity in Ω can be expressed as

$$\epsilon_r(\mathbf{E}(\mathbf{r}), \mathbf{r}) = \begin{cases} \bar{\epsilon}_r(\mathbf{r}), & \mathbf{r} \in \Omega_1 \\ \tilde{\epsilon}_r(\mathbf{E}(\mathbf{r}), \mathbf{r}) = \bar{\epsilon}_r(\mathbf{r}) + \mathcal{L}(\mathbf{E}(\mathbf{r})), & \mathbf{r} \in \Omega_2 \end{cases} \quad (1)$$

where $\mathbf{E}(\mathbf{r})$ is the electric field in the generic point $\mathbf{r}$ and $\mathcal{L}(\cdot)$ denotes a generic scalar operator describing the nonlinear behavior [7].

The electric field satisfies, within Ω , the vector Helmholtz equation

$$\nabla \times \left(\frac{1}{\mu_r} \nabla \times \mathbf{E}(\mathbf{r}) \right) - k^2 \epsilon_r(\mathbf{E}(\mathbf{r}), \mathbf{r}) \mathbf{E}(\mathbf{r}) = 0 \quad (2)$$

with suitable boundary conditions on Γ , being k the free space wavenumber. For the example shown in section III, Γ comprises a part Γ_0 which will be considered perfect electric

conductor, and L parts Γ_l corresponding to the ports, where a modal expansion of the field is enforced [8]. Γ_{12} corresponds to the internal boundary between subdomains Ω_1 and Ω_2 .

In a standard FE Galerkin framework the domain Ω is discretized into a set of non overlapping finite elements and the electric field is approximated as a linear combination of vector basis functions α_n :

$$\mathbf{E}(\mathbf{r}) = \sum_{n=1}^N E_n \alpha_n(\mathbf{r}) \quad (3)$$

Then by testing (2) over Ω with weighting functions α_m , chosen equal to the basis, the following weak form is obtained:

$$\begin{aligned} \int_{\Omega} \nabla \times \alpha_m(\mathbf{r}) \cdot \frac{1}{\mu_r} \nabla \times \mathbf{E}(\mathbf{r}) d\Omega \\ - k^2 \int_{\Omega} \alpha_m(\mathbf{r}) \cdot \varepsilon_r(\mathbf{E}(\mathbf{r}), \mathbf{r}) \mathbf{E}(\mathbf{r}) d\Omega \\ + \oint_{\Gamma} \alpha_m(\mathbf{r}) \cdot \hat{n} \times \frac{1}{\mu_r} \nabla \times \mathbf{E}(\mathbf{r}) d\Gamma = 0 \quad (4) \end{aligned}$$

The multiharmonic dependence in the HBEF method is introduced by approximating the electric field as [5]

$$\hat{\mathbf{E}}(\mathbf{r}, t) = \sum_{p=1}^P \left(\mathbf{E}_p^{(s)}(\mathbf{r}) \sin(p\omega_0 t) + \mathbf{E}_p^{(c)}(\mathbf{r}) \cos(p\omega_0 t) \right) \quad (5)$$

where ω_0 is the fundamental angular frequency and, here and in the following, the hat over a quantity denotes its approximated value given by the truncated Fourier series. The Fourier expansion is devoid of a DC-term as a static field does not represent a solution to (2) for $k = \omega\sqrt{\epsilon_0\mu_0} > 0$. $\mathbf{E}_p^{(s)}$ and $\mathbf{E}_p^{(c)}$ are the harmonic complex amplitudes that expand the field at $p\omega_0$ and hence they must individually satisfy (2) with $k = p\omega_0$. This is done in the FE framework by using the expansion (3) for each coefficient $\mathbf{E}_p^{(s)}$ and $\mathbf{E}_p^{(c)}$.

The scalar relative permittivity can also be approximated as a truncated Fourier series

$$\hat{\varepsilon}_r(\mathbf{E}(\mathbf{r}), \mathbf{r}, t) = \varepsilon_{r_0} + \sum_{g=1}^G \left(\varepsilon_{r_g}^{(s)} \sin(g\omega t) + \varepsilon_{r_g}^{(c)} \cos(g\omega t) \right) \quad (6)$$

and the coefficients of the expansion are given by

$$\begin{aligned} \varepsilon_{r_0} &= \frac{\omega}{2\pi} \int_0^{\frac{2\pi}{\omega}} \varepsilon_r(\mathbf{E}(\mathbf{r}), \mathbf{r}) dt \\ \varepsilon_{r_g}^{(s)} &= \frac{\omega}{\pi} \int_0^{\frac{2\pi}{\omega}} \varepsilon_r(\mathbf{E}(\mathbf{r}), \mathbf{r}) \sin(g\omega t) dt \\ \varepsilon_{r_g}^{(c)} &= \frac{\omega}{\pi} \int_0^{\frac{2\pi}{\omega}} \varepsilon_r(\mathbf{E}(\mathbf{r}), \mathbf{r}) \cos(g\omega t) dt \end{aligned} \quad (7)$$

The orders of the approximations P and G reflects on the accuracy of the solution and must be carefully chosen for the specific device analyzed and for the intensity of the incident field. They can be set automatically in the algorithm by selecting an appropriate convergence criterion on the power associated to the fundamental frequency as the number of harmonic coefficients is increased [9].

A conventional, linear, FE approach discretization of (4) via (3) leads to a system:

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1N} \\ a_{21} & a_{22} & \dots & a_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ a_{N1} & a_{N2} & \dots & a_{NN} \end{bmatrix} \begin{bmatrix} E_1 \\ E_2 \\ \vdots \\ E_N \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_N \end{bmatrix} \quad (8)$$

where a_{ij} are built by assembling the integrals in (4) on an element-by-element basis [8]. In an HBEF approach, $\mathbf{E}(\mathbf{r})$ and $\varepsilon_r(\mathbf{r})$ must be substituted by their respective harmonic approximations $\hat{\mathbf{E}}(\mathbf{r}, t)$ and $\hat{\varepsilon}_r(\mathbf{r}, t)$ in (4). By performing a further testing with weights $\sin(q\omega t)$ or $\cos(q\omega t)$, $q = 1 \dots P$, and integrating over the time period of the fundamental $[0, \frac{2\pi}{\omega}]$, a large linear system is obtained which has formally the same structure as (8) but in which the generic entries a_{mn} , E_n and b_m are themselves matrices and vectors of the form:

$$a_{mn} = \begin{bmatrix} a_{mn}^{(1s1s)} & a_{mn}^{(1s1c)} & \dots & a_{mn}^{(1sPc)} \\ a_{mn}^{(1c1s)} & a_{mn}^{(1c1c)} & \dots & a_{mn}^{(1cPc)} \\ \vdots & \vdots & \ddots & \vdots \\ a_{mn}^{(Pc1s)} & a_{mn}^{(Pc1c)} & \dots & a_{mn}^{(PcPc)} \end{bmatrix} \quad (9)$$

$$E_n = \begin{bmatrix} E_n^{(1s)} & E_n^{(1c)} & \dots & E_n^{(Pc)} \end{bmatrix}^T \quad (10)$$

$$b_m = \begin{bmatrix} b_m^{(1s)} & b_m^{(1c)} & \dots & b_m^{(Pc)} \end{bmatrix}^T \quad (11)$$

The superscripts $(q[s|c]p[s|c])$ of a_{mn} and $(p[s|c])$ of E_n and b_m indicates that the corresponding entry was computed by testing (4) with $[\sin(q\omega t)|\cos(q\omega t)]$ while the corresponding harmonic basis of (5) was $[\sin(p\omega t)|\cos(p\omega t)]$.

The product that occurs in the second integral of (4), between the approximated field and the permittivity, induces coupling between the harmonic components which reflects on non-diagonal subentries of a_{mn} . However, as the PDE we seek to solve is free of first order time derivative, there is no coupling between $\mathbf{E}_n^{(ps)}$ and $\mathbf{E}_n^{(pc)}$. Thus, if either the sine or the cosine harmonic function is considered as the excitation, only the corresponding function is needed to express the multiharmonic behavior of the field, halving in such a way the system dimensions and leading to an NP order system.

The nonlinearity is finally handled via a relaxed iteration. As a starting point a null electric field $[E]^0$ is assumed, computing the pertinent system matrix $[A]^0 = [A_{([E]^0)}]$ and known term $[b]^0 = [b_{([E]^0)}]$ and solving the system $[A]^0[E]^1 = [b]^0$. The newly computed field $[E]^1$ can now be used to update system matrix and known term and perform the following relaxed iteration:

$$[A]^{i-1}[E]^i = [b]^{i-1} \quad (12)$$

with

$$[A]^{i-1} = [A_{(\gamma[E]^{i-1} + (1-\gamma)[E]^{i-2})}] \quad (13)$$

$$[b]^{i-1} = [b_{(\gamma[E]^{i-1} + (1-\gamma)[E]^{i-2})}] \quad (14)$$

with $i \geq 2$ and $\gamma \in (0, 1]$. $\gamma = 1$ gives the standard Picard iteration [7]. Highly nonlinear materials and high intensity

impressed fields may hamper the convergence of the nonlinear iteration, due to oscillation of the solutions around the expected values. Lowering the value of γ has the effect to damp that oscillation, hence rendering the iteration more robust but inevitably reducing the convergence rate [10].

The process is repeated until the relative error between the updated solution and the previous one, in the sense of the Euclidean norm, is less than a prescribed value τ :

$$\frac{\| [E]^i - [E]^{i-1} \|}{\| [E]^i \|} \leq \tau \quad (15)$$

The order of the resulting HBFE system is NP . In practical analyses, this number can become considerable, and, since the solutions have to be refined iteratively building at each step a new linearized system, computing time can become too high.

To enhance the computational efficiency, a substructuring DD approach is here proposed [7], [11]–[13]. The multiharmonic unknowns vector $[E]$ defined by the HBFE method over Ω can be split into two vectors $[E^{(1)}]$ and $[E^{(2)}]$ containing the unknowns belonging to interior points in Ω_1 and Ω_2 , respectively. Unknowns belonging to the boundaries $\Gamma \cup \Gamma_{12}$ are placed in a third vector $[E^{(c)}]$. The HBFE system can then be recast in the following HBDDFE form:

$$\begin{bmatrix} [M^{(1)}] & 0 & [P^{(1)}] \\ 0 & [M^{(2)}]^{i-1} & [P^{(2)}]^{i-1} \\ [Q^{(1)}] & [Q^{(2)}]^{i-1} & [C]^{i-1} \end{bmatrix} \begin{bmatrix} [E^{(1)}]^i \\ [E^{(2)}]^i \\ [E^{(c)}]^i \end{bmatrix} = \begin{bmatrix} [b^{(1)}] \\ [b^{(2)}]^{i-1} \\ [b^{(c)}]^{i-1} \end{bmatrix} \quad (16)$$

where $[M^{(1)}]$ contains the HBFE coefficients of the linear system related to the unknowns $[E^{(1)}]$ fully internal to Ω_1 , hence with null harmonic coupling coefficients. $[M^{(2)}]$ contains coefficients for the nonlinear subdomain Ω_2 . $[P^{(\cdot)}]$ and $[Q^{(\cdot)}]$ represent couplings between the interior unknowns and the ones on the boundaries which are collected in $[E^{(c)}]$.

By using the Schur complement concept, based on a Gauss elimination-type strategy, the boundary unknowns $[E^{(c)}]^i$ of (16), and hence the generalized scattering matrix of the device, can be retrieved [7], [13].

Hence, in order to solve the HBDDFE system and compute the steady state field, every single submatrix is assembled at first sight, then, within the iteration loop, only the submatrices related to Ω_2 and Γ_{12} are to be updated and a recursive system solution performed. The efficiency of this DD technique relies on the fragmentation of the matrices and the subsequent computation of partial solutions. When the number of coefficients related to the nonlinear media is small, noticeable improvements can be achieved.

III. NUMERICAL RESULTS

In this section, the analysis of a millimeter wave passband filter [14] is performed and the computational results are shown. The filter is realized within a standard WR6 waveguide by properly designed metalizations on a dielectric slab, placed in the E-plane (Fig. 2). First order nodal elements are used and the domain discretization leads to 2470 degrees of freedom, 42 of which belong to the nonlinear subdomain Ω_2 , 456 to the boundaries and the remaining 1972 to the linear subdomain

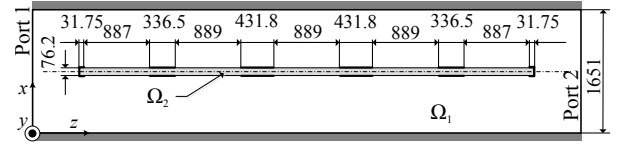


Fig. 2. Cross section (H-plane) of a passband filter realized by placing on the E-plane a dielectric slab partially metalized on both sides (slab is shown in light grey; metal strips - uniform along y are shown as thick black lines) in a WR6 rectangular waveguide [14]. Measures are given in μm .

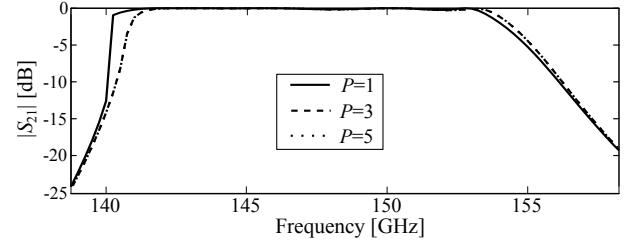


Fig. 3. Behavior of the TE_{10} spectral response as the number of harmonics retained for the HBFE increases ($E_i = 10 \text{ kV/m}$).

Ω_1 . Such a fine discretization is necessary to obtain a good approximation of the field up to the 5th harmonic.

The waveguide walls and the metalizations are considered perfectly conducting. The continuity of the field at the ports is imposed through a modal expansion which, for the symmetry of the problem, exploits only TE_{m0} modes, $m = 2n + 1$, $n \in \mathbb{Z}$. In the following, modal expansion is truncated at $n \leq 4$. The excitation is a TE_{10} mode impinging at Port 1 whose maximum field amplitude is E_i .

The dielectric, enclosed in Ω_2 , presents a Kerr-type nonlinearity such that [7], [15]

$$\tilde{\epsilon}_r(\mathbf{E}(\mathbf{r}), \mathbf{r}) = \bar{\epsilon}_r(\mathbf{r}) + \alpha_2 |\mathbf{E}(\mathbf{r})|^2, \quad \mathbf{r} \in \Omega_2 \quad (17)$$

where $\bar{\epsilon}_r(\mathbf{r}) = 2.1$ and α_2 is chosen $1.625 \cdot 10^{-10} \text{ m}^2/\text{V}^2$ according to [16]. The Kerr-like behavior of the permittivity induces the generation of odd order harmonics, thus the even ones have been neglected in the truncated Fourier expansion. Furthermore, as a sinusoidal excitation is chosen, only $\sin(p\omega t)$ related coefficients of the electric field are retained. For the relaxed iteration stop criterion, $\tau = 10^{-6}$ has been chosen.

As a first analysis, the effect of the total number of harmonics P retained on the filter's spectral response is investigated (Fig. 3). For an incident field amplitude $E_i = 10 \text{ kV/m}$ (corresponding to an incident power equal to 72mW) the Picard iteration proved to converge. It is evident from Fig. 3 that including the 3rd harmonic is crucial for proper evaluation of the device bandwidth.

To better show the nonlinear behavior, the simulations have been repeated with $E_i = 5 \text{ kV/m}$ and $E_i = 15 \text{ kV/m}$ corresponding to 18mW and 162mW of incident power, respectively, which are common power values encountered in D-band applications [17]. The results, compared to those of a linear device ($\alpha_2 = 0$), are shown in Fig. 4 where the computations are carried out with $P = 5$. There is an evident shift of the bandpass range towards lower frequencies as higher power densities are involved. Furthermore the $E_i = 15 \text{ kV/m}$

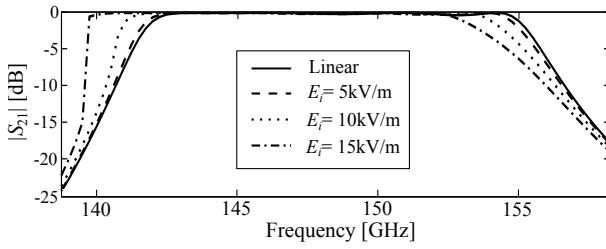


Fig. 4. S_{21} amplitude responses of the nonlinear filter given for several values of E_i compared to the linear case. The field is approximated up to the 5th harmonic.

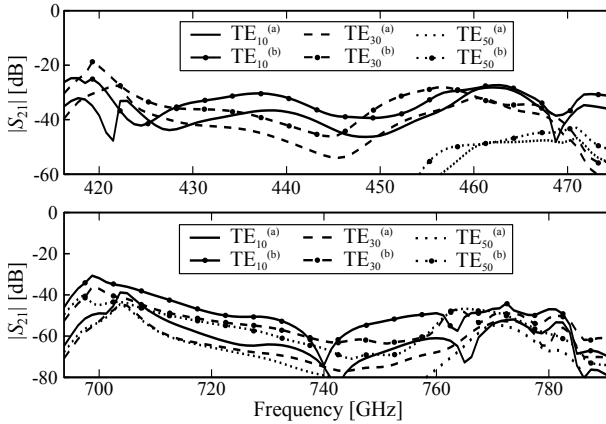


Fig. 5. Spectral responses at 3rd (top) and 5th (bottom) harmonics for the relevant guided modes with an incident TE_{10} of (a) $E_i = 10\text{kV/m}$ and (b) $E_i = 15\text{kV/m}$.

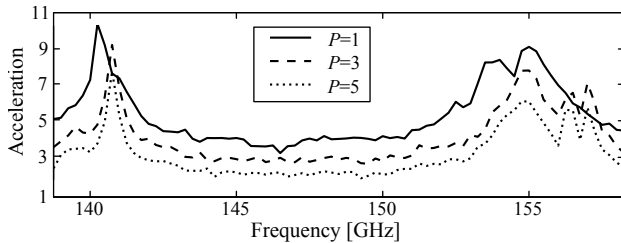


Fig. 6. Acceleration spectrum obtained for an HBFE system of several harmonic orders. The computations are made for $E_i = 10\text{kV/m}$.

did not converge with Picard iteration and required a relaxed $\gamma = 0.1$ iteration. In this case, the abrupt variation of $|S_{21}|$ around 139.5GHz points out that $P = 5$ is not sufficient to accurately analyze the device. Indeed, the power associated to harmonics grows as the incident field increases as shown in Fig. 5.

The efficiency of the DD technique versus the full domain solution can be assessed by its acceleration, that is the ratio between time for a full domain computation - inclusive of nonlinear iterations - and the corresponding DD computation [7]. Fig. 6 presents the acceleration obtained for each frequency point, some frequencies requiring more iterations than others, and for different values of P , having set $E_i = 10\text{kV/m}$. In this case, as stated before, at least a $P = 3$ is required for an accurate solution.

In general, higher order HBDDFE systems have lower acceleration, since the matrices dimensions related to nonlinear

media increase. With a $P = 3$ system, the acceleration varies in the range [2.51, 9.15], averaging to 4.18, and leading to an HBDDFE solution in 24% of the time required by a full domain HBFE solution.

IV. CONCLUSION

An harmonic balance approach to the FE solution of waveguide devices containing nonlinear dielectrics has been presented and the accuracy of the multiharmonic solution discussed. A domain decomposition technique has been proposed to speed-up the computations and the efficiency of the algorithm has been assessed.

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